

Bayesian learning and Monte Carlo Simulations: final project 2026

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1 General Instructions

You are expected to write a short report (about 8–10 pages long; additional details may be included in an appendix) containing the following sections:

- Brief description of the data and of the problem.
- Model specification: describe the model for the data (likelihood) and the prior distribution. You may include a short discussion on the **choice of the hyperparameter** values in the prior (if hyperparameters are present). Be clear about the parametrization and the distributions used, both for the likelihood and for the prior.
- **Posterior analysis and interpretation of the results:** include some plots of the posterior distributions, as well as summary statistics (e.g., posterior means, variances, and credible intervals). Any sensitivity analysis should also be included here; for example, trying different values of the prior variance in the case of linear models. Specify the algorithm used to sample from the posterior distribution, e.g., whether you use BAS or implement MCMC methods using JAGS. Add some details on this point. Do not report the code in the report; it must be uploaded separately.
- If needed: model selection or comparison with alternative models. For example, in regression problems, you may compare your results with those obtained from a linear model when the data are not normally distributed. Alternatively, you may first perform model selection and then analyze the posterior distributions of the selected model(s).
- Any prediction exercise will be appreciated. If covariates are present, you may split the data into two parts, fit the model using the first part of the data, and use the second part for an out-of-sample prediction exercise. Similar analyses may also be carried out for time series models.
- Final comments and conclusions.

Additional suggestions

- You may include **an appendix with additional analyses** (such as autocorrelation plots, trace plots, or other diagnostic tools/tests).

- The code should be provided in separate R file(s), ready to be run if needed, and should include short comments to make the analysis understandable.

Mandatory: the R file(s) and the the Project (pdf format) need to be submitted at least 3 day before the examination.

2 Final Examination

- **Presentation format:** The project presentation should be delivered using slides and should last approximately 20–25 minutes. Each member of the group is expected to actively participate in the presentation and discuss a specific part of the project. This is intended to ensure a comprehensive and well-balanced presentation that reflects the collective work of the entire group.
- **Questions from the examining committee:** During the discussion, the examining committee may ask questions related both to the project and to topics covered in the course. Such questions may also concern course topics that are not directly addressed in the project itself. These questions may be directed to one or more members of the group. All group members are therefore expected to be well prepared and familiar with the project, the related concepts, and the implementation details, including the code used in the analysis.

3 Datasets

All datasets (together with some additional comments and supporting material) are available on the WeBeep page.

Before choosing a dataset, please read the corresponding `Project2026.Rmd` file on WeBeep, which contains additional information and some preliminary data visualizations.

1. Bike Sharing Data
2. Yacht Hydrodynamic Resistance
3. Airline Customer Satisfaction
4. Bank Marketing
5. Student Performance Data
6. JFK Passengers
7. GDP, Inflation & Unemployment (US)
8. ARPA: Air Quality
9. SPF
10. Single-cell Growth Measurements
11. Hourly City Load Factor
12. Capacitor Accelerated Life Tests

3.1 Bike Sharing Data

Source. <https://www.kaggle.com/code/juniorbueno/rental-bikes/notebook>

Short description. The data contains the number of casual/registered users in bike sharing systems and various additional covariates (related to the weather) as well information on days/month/year.

Name of the file. `bike.csv`

Task. Fit a regression model to predict the total number of rented bikes (or, alternatively, the number of casual or registered users). Since the counts are relatively large, you may consider using the logarithm of the counts together with a Gaussian model. Discuss the impact of the different variables, paying particular attention to the presence of several categorical covariates. Perform model selection and include a prediction exercise.

3.2 Yacht Hydrodynamic Resistance

Source. UCI Machine Learning Repository,
<https://archive.ics.uci.edu/dataset/243/yacht+hydrodynamics>.

Original record: Gerritsma, Onnink and Versluis, *Geometry, resistance and stability of the Delft Systematic Yacht Hull Series*, 1981.

Name of the file. `yacht_hydro.csv`

Short description. The dataset describes 22 different hull forms (sailing yachts), each tested at multiple Froude numbers in a towing tank, producing 308 hull-form $\times$ Froude-number rows. The variables are:

- **LCB:** longitudinal position of the center of buoyancy (% length);
- **prismatic:** prismatic coefficient (dimensionless);
- **L_disp:** length-displacement ratio;
- **beam_draught:** beam-draught ratio;
- **L_beam:** length-beam ratio;
- **Froude:** Froude number (dimensionless, $Fn = v/\sqrt{gL}$), in $[0.125, 0.45]$; the dominant predictor;
- **resistance:** residuary resistance per unit weight of displacement (target variable).

Task. The physics of the problem says that for sailing yachts the residuary resistance follows a power law of the Froude number with exponent close to 4 in the bulk regime. A good Bayesian project explores this:

1. **Log-log regression.** Fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_k \beta_k \text{hull}_k + \varepsilon.$$

Verify that $\hat{\beta}_F \approx 4$.

2. **BAS / BMA.** Use `bas.lm` to enumerate the 2^5 submodels over the five hull-geometry covariates and identify which of them (LCB, `prismatic`, `L/disp`, `B/draught`, `L/beam`) carry signal once `Froude` is in the model.
3. **Correct for heteroscedasticity.** Train the nonlinear model

$$\text{resistance} = k \text{Froude}^{\gamma_F} \exp\left\{\sum_k \beta_k \text{hull}_k\right\} + \varepsilon,$$

where only the form factors that appear relevant in the model selection phase are kept for the nonlinear modelling. Compare the results with the log-log regression and discuss the impact of the heteroscedasticity correction on the inference on γ_F and on the hull-geometry covariates.

4. Plot the posterior predictive curves and compare them with or without the treatment of the heteroscedasticity, and compare them with the data.

3.3 Airline Customer Satisfaction

Source. <https://www.kaggle.com/datasets/raminhuseyn/airline-customer-satisfaction>

Name of the file. `airline_sub.csv` (full dataset: `Airline_customer_satisfaction.csv`)

Short description. The dataset provides information on customer satisfaction levels for an undisclosed airline company. Although the name of the airline is not provided, the dataset contains a rich set of variables, with a total of 22 columns.

For convenience, we extracted a subset containing 1000 customers from the original dataset.

Task. Perform a suitable regression analysis for the categorical response variable `Satisfaction`. Identify the variables that are most strongly associated with customer satisfaction. A prediction exercise is required. Variable selection methods may also be discussed if desired.

It is recommended to start using only a subset of the covariates, for example:

`Age`, `SeatComfort`, `FlightDistance`, `Class`, `DepartureDelay`, `ArrivalDelay`.

3.4 Bank Marketing

Source. <https://archive.ics.uci.edu/dataset/222/bank+marketing>

Name of the file. `bank.csv`

Short description. The dataset is related to direct marketing campaigns conducted by a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact with the same client was required in order to determine whether the client would subscribe to a bank term deposit (`yes`) or not (`no`).

Task. Perform a suitable regression analysis for the categorical response variable corresponding to the subscription of a bank term deposit. Identify the variables that are most strongly associated with the response. A prediction exercise is required. Variable selection methods may also be discussed if desired.

Since the dataset contains many categorical covariates, you may choose to work only with a subset of them.

3.5 Student Performance Data

Source. <https://archive.ics.uci.edu/dataset/320/student+performance>

Name of the file. [student-mat.csv](#)

Short description. This dataset concerns student achievement in secondary education at two Portuguese schools. The data include student grades together with demographic, social, and school-related features. The dataset was collected using both school reports and questionnaires.

Task. Fit a regression model to predict the final grade. Pay attention to the fact that the dataset contains many categorical variables. You may decide to exclude some covariates a priori in order to simplify the regression model.

Hint. The final grade is not a categorical variable. Use a suitable discrete distribution with support on the range of the grades, and include the covariates in its parameters through appropriate link functions.

3.6 JFK Passengers

Source. https://github.com/alan-turing-institute/TCPD/tree/master/datasets/jfk_passengers

Name of the file. [JFK.csv](#)

Short description. The Port Authority collects monthly data on domestic and international passengers, cargo, flights, and aircraft equipment types from each carrier operating at PANYNJ airports.

Task. This dataset can be analyzed using a change-point model with different means before and after the change point, possibly including a linear drift after the change point. A basic model is

$$y_t = \mu_t + \epsilon_t,$$

where

$$\epsilon_t \sim \mathcal{N}(0, \sigma_t^2),$$

and

$$\mu_t = \mu_1, \quad \sigma_t = \sigma_1 \quad \text{if } t \leq \tau,$$

while

$$\mu_t = \mu_2, \quad \sigma_t = \sigma_2 \quad \text{if } t > \tau.$$

Here τ is an unknown change-point parameter.

More complex change-point models may also be considered and discussed. Time series models are encouraged; for example, one may include different trends and different variances before and after the change point.

3.7 US GDP, Inflation & Unemployment

Source. <https://fred.stlouisfed.org/series/HSN1F>

Name of the file. [GDP-CPI-UNE.csv](#)

Short description. The dataset consists of three time series:

- Gross Domestic Product (GDP): real gross domestic product per capita (A939RX0Q048SBEA) [percent change from one year ago];
- Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPI-AUCSL) [percent change from one year ago];
- Unemployment Level (UNEMPLOY) [percent change from one year ago].

More information is available on the website.

Task. Fit suitable time series models for one or more of the series, analyzed separately. For example, you may consider AR, MA, ARMA, or GARCH models. If you use more than one model, compare the models using suitable information criteria, such as BIC, DIC, or WAIC.

You may also consider multivariate time series models, such as a simple VAR(1) model. In this case, please ask the instructor for further information.

The website also provides information on U.S. recession years. You may try to fit a switching model with two regimes: recession and non-recession.

3.8 ARPA: Air Quality

Source. ARPA Lombardia Open Data Portal <https://www.dati.lombardia.it/stories/s/auv9-c2sj>

Name of the file. [ARPA.csv](#)

Short description. Air quality data collected at the Milano Pascal Città Studi monitoring station (station ID: 705):

- PM10: PM10 daily average concentration, $\mu\text{g}/\text{m}^3$;
- BioAz: NOx daily average concentration, $\mu\text{g}/\text{m}^3$;
- precipitazioni: daily precipitation, mm;
- temper: daily average temperature, degrees Celsius;
- vento: daily average wind speed, m/s;
- umidita: daily average air humidity.

Task. Choose one of the following tasks.

Task 1. Perform a suitable regression analysis for either:

- PM10 (integer-valued response), or
- BioAz (continuous response),

using the remaining variables as covariates.

Note that the dataset also includes the day of the year (**giorno**), which may be useful for capturing seasonal effects.

Task 2. Consider a time series model for either PM10 or BioAz. You may:

- use a simple model without exogenous covariates, or

- include the other variables as exogenous covariates.

Again, note that the dataset also includes the day of the year (**giorno**), which may help account for seasonality.

3.9 SPF

Source. <https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/survey-of-professional-forecasters>

Name of the file. **SPF_GDP.csv**

Short description. The Survey of Professional Forecasters (SPF) collects forecasts of macroeconomic variables. Both the U.S. SPF and the ECB-SPF ask forecasters to provide point forecasts and density forecasts. Density forecasts are reported in the form of histograms over a set of intervals specified in the survey instrument.

Additional information about the dataset is provided in the notebook **SPF**.

Task.

Investigate whether individual uncertainty appears to be associated mainly with respondent-specific effects, while point forecasts (e.g., mean or median forecasts) are more strongly affected by time-period effects.

In other words, the goal is to study whether there are substantial differences across forecasters in the confidence attached to their predictions, while each forecaster's confidence evolves only slowly over time.

Consider only observations corresponding to a fixed forecast horizon (e.g., horizon 1, 2, etc.). If the variable of interest is denoted by $y_{f,t}$ (for example, the forecast mean or $y = \log(\sigma)$), start with the model

$$y_{f,t} = \alpha_t + \beta_f + \epsilon_{f,t},$$

where α_t represents the time effect and β_f the forecaster effect.

Compare variables related to point forecasts (e.g., mean or median forecasts) with variables related to uncertainty (e.g., variance or probability assigned to the modal interval).

More advanced models may also be considered. For example:

- models with time-varying error variances;
- mixture models in which forecasters are grouped into latent blocks, and responses depend only on the block membership.

Please ask the instructor for additional explanations if you are interested in this project.

3.10 Single-cell Growth Measurements

Source. Time-lapse microscopy experiments on a bacterial cell line (P5ori). Kennard, A. S., Osella, M. C., Javer, A., Grilli, J., Nghe, P., Tans, S., Cicuta, P., Cosentino Lagomarsino, M. C. (2015). *Individuality and universality in the growth-division laws of single E. coli cells*. Physical Review E, 93, 012408 (ask the teacher for further details and/or additional data from the same study).

Name of the file. `cells_wide.csv`

Short description. The dataset contains single-cell morphological measurements collected from time-lapse microscopy experiments on a bacterial cell line (P5ori) grown under two different culture conditions: Glc (glucose) and RDM (Rich Defined Medium). Each row corresponds to one observation of an individual cell in a given frame of the experiment.

The columns are:

- the first column is an index uniquely identifying the cell;
- `medium`: identifies the growth medium, either Glc or RDM;
- `t0`, `L0`: time and length at birth (first frame of the cell's life);
- `t1`, `L1`, ..., `t61`, `L61`: time and length at subsequent frames until the cell divides (last frame of the cell's life).

The final frame is not the same for all cells, hence the presence of trailing NA values.

Task. For each cell, the length L typically grows exponentially between two divisions, that is

$$L(t) = L(t_0)e^{\beta(t-t_0)}$$

where $L(t_0)$ is the length at birth time t_0 and $L(t)$ is the length at time t . Fit a model of the form

$$\log\left(\frac{L_i(t)}{L_i(t_0)}\right) = \beta_i(t - t_0) + \epsilon_{i,t}$$

with independent Gaussian errors $\epsilon_{i,t}$ and cell-specific coefficients β_i . Study the distribution of the growth rates β_i across conditions. Use hierarchical modeling to share information across cells and obtain more precise estimates of the growth rates. Discuss whether the two media Glc and RDM lead to significantly different cell sizes and/or growth rates.

3.11 Hourly City Load Factor

Source. Salam & El Hibaoui, *Power Consumption of Tetouan City*, UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city>.

Name of the file. `hourly_load_factor.csv`

Short description. The original release records the active power consumed by the three distribution zones of Tetouan (Morocco) during 2017, sampled every ten minutes by the SCADA system of the local utility. The CSV we share is a derived, pre-cleaned version. For each (day, hour) of the year we have computed the hourly mean of the *city-wide* active power

$$\overline{P}(d, h) = \text{Zone1}(d, h) + \text{Zone2}(d, h) + \text{Zone3}(d, h)$$

from the original 10-minute readings, and then the *hourly load factor*

$$\text{LF}(d, h) = \frac{\overline{P}(d, h)}{\max_{h'} \overline{P}(d, h')},$$

i.e. the ratio between the average load in hour h and the highest hourly average of the same day. By construction $\text{LF}(d, h) \in [0, 1]$: the day-peak hour gets exactly 1 and the deepest

off-peak hour gets the smallest value. From the full $365 \times 24 \approx 8700$ (day, hour) cells we have taken a uniform random subsample of 2000 rows and clamped the values to $(10^{-3}, 1 - 10^{-3})$ so that the Beta log-density is finite at the boundary.

The variables are:

- **date**: calendar date (M/D/YYYY);
- **hour**: hour of the day, integer in $\{0, \dots, 23\}$;
- **load_factor**: hourly load factor as defined above, in $(10^{-3}, 1 - 10^{-3})$.

Task.

1. Fit a single Beta model and a Beta mixture with $H = 1, 2, 3, 4, 5$ components, performing model selection via WAIC.
2. Overlay the posterior predictive density on the empirical histogram and verify that the modes are recovered.
3. **Post-hoc exploration.** Assign each observation to its maximum-a-posteriori component, then plot the empirical fraction of observations in each component as a function of **hour**.
4. **Advanced.** Fit a second model which performs classification on **hour** using the component assignment as response variable, and plot the predicted probability of each component as a function of **hour**. Do the two plots match? Discuss the results. (Details for this step are presented in the additional details notebook.) The component assignment is predicted using a multinomial logistic regression with the Fourier-transformed **hour** as covariate, i.e. with predictors $\sin(k 2\pi h/24)$ and $\cos(k 2\pi h/24)$ to capture the periodicity of the day.

3.12 Capacitor Accelerated Life Tests

Source. Dataset `capacitor` from the R package `survival`.

Name of the file. `capacitor` (available via `data(capacitor)` after loading the package with `library(survival)`)

Short description. The dataset contains the results of 64 accelerated life tests on capacitors. The experiment follows a 4×2 factorial design involving two stress variables:

- **voltage**: the operating voltage during the test (4 levels);
- **temperature**: the operating temperature during the test, in degrees Celsius (2 levels; to be converted to Kelvin for the physical model);
- **time**: observed failure time or censoring time, in hours;
- **status**: censoring indicator (1 = failure observed, 0 = right-censored).

Task. A commonly used physical hazard model for capacitor failure is

$$h(t \mid V, T) \propto V^\gamma \exp(-\beta/T),$$

with T measured in Kelvin. This corresponds to a regression on $\log(V)$ and $1/T$.

1. Fit a proportional hazards model with linear predictor involving $\log(V)$ and $1/T$, and estimate the hazard ratio associated with each variable.
2. Choose an appropriate parametric form for the baseline hazard, for example exponential, Weibull, or log-normal, and compare alternatives using information criteria such as DIC/WAIC, or posterior predictive checks.
3. Extrapolate to typical *operating* conditions, namely 50°C and 200 V, which lie *outside* the experimental design, and obtain a posterior cloud or credible band for the survival function

$$S(t \mid V = 200, T = 323.15).$$

4. (More advanced.) Compute the posterior distribution of the *B10 lifetime*, i.e. the time at which 10% of the components have failed under operating conditions, namely 50°C and 200 V:

$$B_{10} = S^{-1}(0.9 \mid V = 200, T = 323.15).$$